

1) Dockerfile with Ubuntu OS and dependencies

created Dockerfile and python file

```
Dockerfile X app.py
Q2_Docker > Dockerfile > ...
  Click here to ask Blackbox to help you code faster
1  # Use an official Ubuntu runtime as the base image
2  FROM ubuntu:latest
3
4  # Update the system and install Python3 and pip
5  RUN apt-get update -y && apt-get install -y python3 python3-pip
6
7  # Setting working directory in the container
8  WORKDIR /app
9
10
11 # Copy the current directory contents into the container at /app
12 COPY . /app
13
14 CMD ["python", "app.py"]
15
```

2) Python Script (app.py)

print("Hello world")

3) Docker image

```
hpcap@hpcap-DIT400TR-55L:~/Desktop/240340141014/Q2_Docker$ sudo docker build -t python-app .
[+] Building 58.4s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 863B
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 918B
=> CACHED [1/4] FROM docker.io/library/ubuntu:latest
=> [2/4] RUN apt-get update -y && apt-get install -y python3 python3-pip
=> [3/4] WORKDIR /app
=> [4/4] COPY . /app
=> exporting to image
=> => exporting layers
=> writing image sha256:b68179f8f85251f49ea3a3eaa51b7791cf628731b0e099932a260e049a0e0effd
=> naming to docker.io/library/python-app

What's Next?
View a summary of image vulnerabilities and recommendations -> docker scout quickview
hpcap@hpcap-DIT400TR-55L:~/Desktop/240340141014/Q2_Docker$
```

```
hpcap@hpcap-DIT400TR-55L:~/Desktop/240340141014/Q2_Docker$ sudo docker image list
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
python-app          latest      b68179f8f852  3 minutes ago  490MB
```

4) Deploy container

```
hpcap@hpcap-DIT400TR-55L:~/Desktop/240340141014/Q2_Docker$ sudo docker run -d -p 5000:5000 python-app
51d05670699f2323e05a1c82782277194a3d496c4176db1f790284d574c6653
```

5) verify container is running

using docker ps we check docker running state